THINK UNLIMITED

BOARDS 2026/27 CBSE

CLASS 10

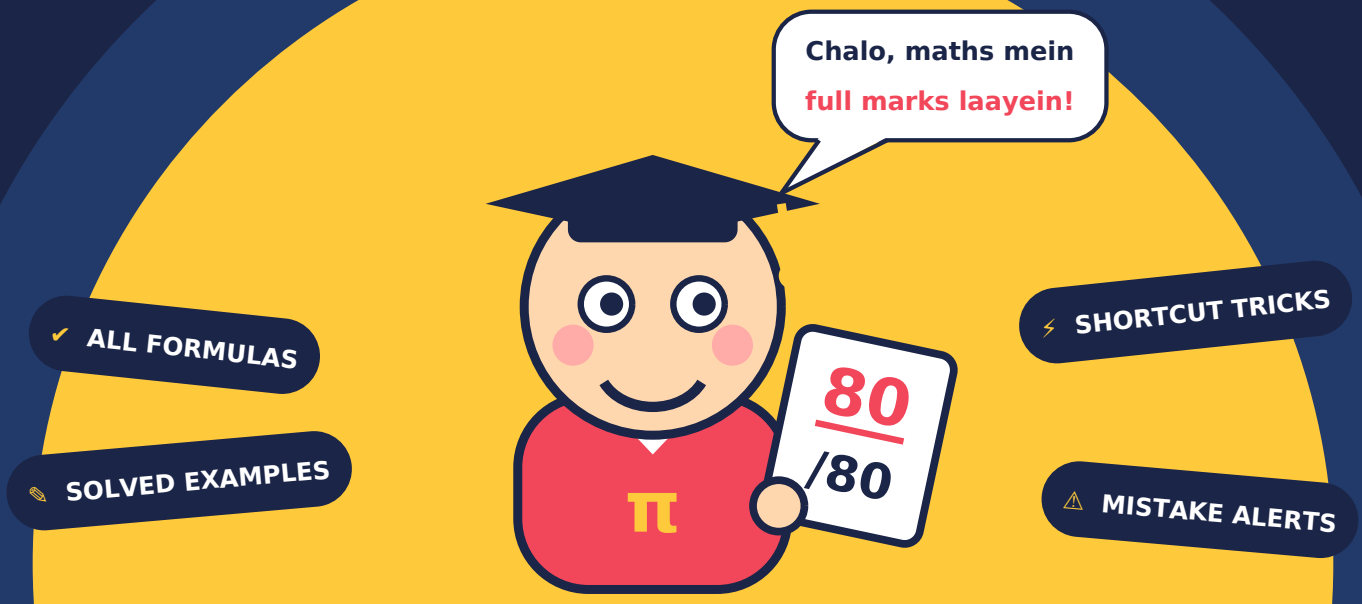
MATHS MADE EASY

All formulas + tricks + examples

14 chapters • 26 pages

Maths se darr? Ab nahi.

CBSE BOARD EXAM EDITION • 2026/27



By Think Unlimited | [Instagram / Telegram handle]

FORMULA • TRICK • EXAMPLE • MOCK TEST — SAB EK HI JAGAH

START HERE

Hey! Pehle yeh padho

Read this first | 2 minutes

14	26	80	1
CHAPTERS	PAGES	MARKS PAPER	BOOK, SAB KUCH

★ MESSAGE

Maths **ratne** ka subject nahi hai. **Formula samjho, 5 sawal lagao, bas.**

Is book mein har formula ke saath likha hai **kab use karna hai**.

→ HOW TO STUDY WITH THIS BOOK

1. Ek chapter padho (10 minutes).
2. Book band karo, formulas copy mein likho.
3. NCERT ke 5 sawal solve karo.
4. Last mein **“Don’t do this”** padho. Yahi marks bachata hai.

◆ WHICH CHAPTERS CARRY THE MOST MARKS? (OUT OF 80)

Unit	Marks	Difficulty
Algebra	20	Medium
Geometry	15	Medium
Trigonometry	12	Easy after practice
Statistics + Probability	11	Easy
Mensuration	10	Easy (just formulas)
Number Systems	6	Easy
Coordinate Geometry	6	Easy

✓ TIP

Start with the **easy chapters**. Confidence builds fast.

CONTENTS

Contents + Tracker

14 chapters | Tick karte jao



→ KYA MILEGA IS BOOK MEIN

Ch	Chapter	Page	Marks	Level	Padha Practice
1	Real Numbers	4	6	Easy	☐ ☐
2	Polynomials	5	20	Medium	☐ ☐
3	Pair of Linear Equations	6	20	Medium	☐ ☐
4	Quadratic Equations	7	20	Medium	☐ ☐
5	Arithmetic Progressions	8	20	Medium	☐ ☐
6	Triangles	9	15	Medium	☐ ☐
7	Coordinate Geometry	10	6	Easy	☐ ☐
8	Trigonometry + Identities	11–12	12	Easy after practice	☐ ☐
9	Applications of Trigonometry	13	12	Easy after practice	☐ ☐
10	Circles	14	15	Medium	☐ ☐
11	Areas Related to Circles	15	10	Easy	☐ ☐
12	Surface Areas & Volumes	16–17	10	Easy	☐ ☐
13	Statistics	18	11	Easy	☐ ☐
14	Probability	19	11	Easy	☐ ☐

✓ 7-DIN PLAN

Roz **2 chapters** padho + NCERT ke 5 sawal. **7 din mein poori book khatam!** Har chapter ke baad ☑ tick lagao.

★ BONUS PAGES

- **Pages 20–21:** Formula Cheat Sheet (poori book ke formulas ek jagah)
- **Page 22:** Squares, cubes aur roots ki speed table
- **Pages 23–24:** Exam-Day Page + Self-Test
- **Pages 25–26:** Mock Test (30 minutes) + Answers

CHAPTER 01

01 Real Numbers

Unit: Number Systems | 6 marks | Difficulty: Easy



★ SIMPLE IDEA

Break every number into **prime factors**. Everything else follows.

Σ FORMULAS

- **HCF × LCM = Number 1 × Number 2** → use it when HCF or LCM is missing
- **HCF** = common primes, **smallest** power
- **LCM** = all primes, **biggest** power
- **Terminating decimal** if the denominator has only 2 and 5 as factors
- √2, √3, √5 are **irrational**

⚡ TRICK

- HCF is the “common” part. LCM is the “everything” part.

✎ EXAMPLE

HCF and LCM of 12 and 18

$12 = 2^2 \times 3$ and $18 = 2 \times 3^2$

HCF = $2 \times 3 = \mathbf{6}$

LCM = $2^2 \times 3^2 = \mathbf{36}$

⚠ DON'T DO THIS • MARKS BACHAO!

Don’t use HCF × LCM = product for **three numbers**. It works for two only.

✔ QUICK CHECK

HCF of 8 and 12? **(Answer: 4)**

CHAPTER 02

02 Polynomials

Unit: Algebra | 20 marks | Difficulty: Medium



★ SIMPLE IDEA

Zeros are the x values where the graph touches the **x-axis**.

Σ FORMULAS (for $ax^2 + bx + c$)

- **Sum of zeros = −b/a**
- **Product of zeros = c/a**
- Make the polynomial: x^2 **− (sum)x + (product)**
- Number of zeros ≤ degree of the polynomial

⚡ TRICK

- Sum has a minus. Product doesn't. Say it aloud: **"Sum minus, product plain."**

✎ EXAMPLE

Quadratic with zeros 2 and −3

Sum = 2 + (−3) = −1

Product = 2 × (−3) = −6

Polynomial = $x^2 + x - 6$

⚠ DON'T DO THIS • MARKS BACHAO!

Don't forget the **minus sign** in −b/a.

✔ QUICK CHECK

Sum and product of zeros of $x^2 - 5x + 6$? **(Answer: 5 and 6)**

CHAPTER 03

03 Pair of Linear Equations

Unit: Algebra | 20 marks | Difficulty: Medium



★ SIMPLE IDEA

Two lines can cross **once**, **never**, or lie **on top of each other**.

◆ THE RATIO RULE ($a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$)

If...	Then...
$a_1/a_2 \neq b_1/b_2$	**One solution**
$a_1/a_2 = b_1/b_2 = c_1/c_2$	**Infinite solutions**
$a_1/a_2 = b_1/b_2 \neq c_1/c_2$	**No solution**

⚡ SHORTCUT TRICKS

- Same or opposite numbers in front of x or y? → **Elimination**
- One variable has coefficient 1? → **Substitution**
- Memory: **“All equal = many, two equal = none.”**

✎ EXAMPLE

x + y = 7 and x − y = 1

Add both: 2x = 8, so **x = 4**

Put in x + y = 7: **y = 3**

Check: 4 − 3 = 1 ✔

⚠ DON'T DO THIS • MARKS BACHAO!

In word problems, always begin with **“Let the ... be x and y”**. Always check your answer.

✔ QUICK CHECK

x + y = 10 and x − y = 2? **(Answer: x = 6, y = 4)**

CHAPTER 04

04 Quadratic Equations

Unit: Algebra | 20 marks | Difficulty: Medium



★ SIMPLE IDEA

The discriminant **D** tells you how many real roots exist.

Σ FORMULAS ($ax^2 + bx + c = 0$)

- **$D = b^2 - 4ac$**
- **$x = (-b \pm \sqrt{D}) / 2a$** → use when factorisation is hard
- $D > 0$ → **2 different** real roots
- $D = 0$ → **2 equal** real roots
- $D < 0$ → **no** real roots

⚡ SHORTCUT TRICKS

- Splitting the middle term: find two numbers with **product = a × c** and **sum = b**
- If **a + b + c = 0**, one root is 1
- “Equal roots” in a question? → put **D = 0**

✎ EXAMPLE

$x^2 - 5x + 6 = 0$

Need two numbers: product 6, sum -5 → -2 and -3

$(x - 2)(x - 3) = 0$

x = 2 or x = 3

⚠ DON'T DO THIS • MARKS BACHAO!

In word problems, **reject negative** lengths, ages, or speeds. Write the final conclusion line.

✔ QUICK CHECK

For what k does $x^2 + kx + 9 = 0$ have equal roots? **(Answer: k = ±6)**

CHAPTER 05

05 Arithmetic Progressions (AP)

Unit: Algebra | 20 marks | Difficulty: Medium



★ SIMPLE IDEA

A list where the gap between numbers stays the same. **2, 5, 8, 11...** (gap = 3)

Σ FORMULAS

- **$d = a_2 - a_1$**
- nth term: **$a_n = a + (n - 1)d$** → use for "find the 10th term" or "which term is..."
- Sum: **$S_n = n/2\ [2a + (n - 1)d]$** → use when the last term is **not** given
- Sum: **$S_n = n/2\ (a + l)$** → use when the last term **is** given

⚡ SHORTCUT TRICKS

- 3 numbers in AP: take **a − d, a, a + d**
- 4 numbers in AP: take **a − 3d, a − d, a + d, a + 3d**
- "Which term is 50?" → put $a_n = 50$ and find n

✎ EXAMPLE

10th term of 2, 5, 8...

a = 2, d = 3, n = 10

$a_{10} = 2 + 9 \times 3 =$ **29**

⚠ DON'T DO THIS • MARKS BACHAO!

n must be a **positive whole number**. If n comes out as a fraction, that number is not in the AP.

✔ QUICK CHECK

Sum of the first 10 natural numbers? **(Answer: 55)**

CHAPTER 06

06 Triangles

Unit: Geometry | 15 marks | Difficulty: Medium



★ SIMPLE IDEA

Same shape, different size means **similar triangles**. Sides stay in proportion.

Σ RULES

- Similarity tests: **AA, SSS, SAS**
- **BPT**: a line parallel to one side divides the other two sides in the same ratio
- **Area ratio = (side ratio)^2**
- **Pythagoras:** Hypotenuse^2 = Base^2 + Perpendicular^2

⚡ SHORTCUT TRICKS

- Learn these right triangles: **3-4-5, 5-12-13, 8-15-17, 7-24-25**
- “Parallel line inside a triangle” → think **BPT** or **AA**

✎ EXAMPLE

Sides of two similar triangles are in ratio 2:3. Area ratio?
Square the ratio
$2^2 : 3^2 =$ **4 : 9**

⚠ DON'T DO THIS • MARKS BACHAO!

Write the **reason beside every step** (“By BPT”, “By AA”). Match the vertices in order.

✔ QUICK CHECK

Right triangle with sides 6 and 8. Hypotenuse? **(Answer: 10)**

CHAPTER 07

07 Coordinate Geometry

Unit: Coordinate Geometry | 6 marks | Difficulty: Easy



★ SIMPLE IDEA

Points on a graph and a few formulas to work with them.

Σ FORMULAS

- **Distance** = $\sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$
- **Midpoint** = $((x_1 + x_2)/2 , (y_1 + y_2)/2)$
- **Section formula** (ratio m:n) = $((mx_2 + nx_1)/(m + n) , (my_2 + ny_1)/(m + n))$
- **Area of triangle** = $\frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$
- Area = 0 → points are in a **straight line**

⚡ TRICK: IDENTIFY SHAPES BY SIDES

- **Rhombus:** all sides equal
- **Square:** all sides equal + diagonals equal
- **Rectangle:** opposite sides equal + diagonals equal
- **Parallelogram:** both diagonals share the same midpoint

✎ EXAMPLE

Distance between (2, 3) and (5, 7)

Differences: 5 − 2 = 3 and 7 − 3 = 4

√(9 + 16) = **5**

⚠ DON'T DO THIS • MARKS BACHAO!

Put brackets around **negative numbers**. Don’t swap x and y.

✔ QUICK CHECK

Midpoint of (2, 4) and (6, 8)? **(Answer: (4, 6))**

CHAPTER 08 • PART 1

08 Trigonometry Basics

Unit: Trigonometry | 12 marks | Difficulty: Easy after practice



★ SIMPLE IDEA

Trigonometry is just **ratios of sides** in a right triangle.

Σ FORMULAS

- **sin θ = P/H, cos θ = B/H, tan θ = P/B**
- cosec θ = H/P, sec θ = H/B, cot θ = B/P

◆ MEMORY LINE

“Pandit Badri Prasad Har Har Bole” → sin = P/H, cos = B/H, tan = P/B

◆ VALUE TABLE

θ	0°	30°	45°	60°	90°
sin	0	1/2	1/√2	√3/2	1
cos	1	√3/2	1/√2	1/2	0
tan	0	1/√3	1	√3	not defined

⚡ TRICK

- sin goes **√0/2, √1/2, √2/2, √3/2, √4/2**. cos is the same list in **reverse order**.

✎ EXAMPLE

If sin θ = 3/5, find cos θ

P = 3, H = 5, so B = 4 (3-4-5 triangle)

cos θ = B/H = **4/5**

⚠ DON'T DO THIS • MARKS BACHAO!

sin and cos are **never more than 1**. sin θ is one thing, not sin × θ.

✔ QUICK CHECK

Value of sin 30° + cos 60°? **(Answer: 1)**

CHAPTER 08 • PART 2

08 Trigonometric Identities

Unit: Trigonometry | 12 marks | Difficulty: Easy after practice



★ SIMPLE IDEA

An identity is **always true**, for any angle.

Σ THE 3 BIG IDENTITIES

- $\sin^2\theta + \cos^2\theta = 1$
- $1 + \tan^2\theta = \sec^2\theta$
- $1 + \cot^2\theta = \text{cosec}^2\theta$
- Complementary angles: $\sin(90° - \theta) = \cos\theta$, $\tan(90° - \theta) = \cot\theta$, $\sec(90° - \theta) = \text{cosec}\,\theta$

◆ HOW TO PROVE ANY IDENTITY (4 STEPS)

1. Pick **one side only** (usually the left)
2. Change everything to **sin and cos**
3. Use an identity or take the **LCM**
4. Reach the other side and write **“Hence proved”**

✎ EXAMPLE

Prove $(1 - \sin^2\theta)\sec^2\theta = 1$

$1 - \sin^2\theta = \cos^2\theta$

$\cos^2\theta \times \sec^2\theta = \cos^2\theta \times (1/\cos^2\theta) =$ **1** ✔

⚠ DON'T DO THIS • MARKS BACHAO!

Never move terms across the “=” sign in a proof. **Work on one side only.**

✔ QUICK CHECK

Simplify $\sin^2\theta + \cos^2\theta + \tan^2\theta$. **(Answer: $1 + \tan^2\theta = \sec^2\theta$)**

CHAPTER 09

09 Applications of Trigonometry

Unit: Trigonometry | 12 marks | Difficulty: Easy after practice



★ SIMPLE IDEA

Find heights and distances using **tan**.

Σ KEY POINTS

- **Angle of elevation:** looking **up** from the horizontal
- **Angle of depression:** looking **down** from the horizontal
- Both angles are measured from the **horizontal line**
- **tan θ = height / distance** (use this most often)

◆ STEPS FOR ANY PROBLEM

1. Draw a **neat figure**
2. Mark the angle and all known values
3. Pick sin, cos, or tan (usually **tan**)
4. Put in the value (30°, 45°, 60°) and solve

Trick: At 45°, **height = distance**.

✎ EXAMPLE

A tower is seen at 60° from 30 m away. Height?

tan 60° = height / 30

height = 30 × √3 = **30√3 m**

⚠ DON'T DO THIS • MARKS BACHAO!

Draw the figure carefully. A messy figure gives a **wrong equation**.

✔ QUICK CHECK

Angle of elevation is 45° from 20 m away. Height? **(Answer: 20 m)**

CHAPTER 10

10 Circles

Unit: Geometry | 15 marks | Difficulty: Medium



★ SIMPLE IDEA

A tangent touches the circle at **exactly one point**.

Σ FACTS

- Tangent ⊥ radius at the point of contact
- Tangents from an outside point are **equal** (PA = PB)
- **Tangent length = $\sqrt{(d^2 - r^2)}$** (d = distance from the centre)

⚡ SHORTCUT TRICKS

- Tangent + radius + line to centre = **right triangle** → use Pythagoras
- Quadrilateral around a circle: **AB + CD = AD + BC**

✎ EXAMPLE

A point is 13 cm from the centre and the radius is 5 cm. Tangent length?

$\sqrt{(13^2 - 5^2)} = \sqrt{(169 - 25)} = \sqrt{144}$

= **12 cm**

⚠ DON'T DO THIS • MARKS BACHAO!

In proofs, write **“Radius ⊥ tangent”** as the reason.

✔ QUICK CHECK

d = 10, r = 6. Tangent length? **(Answer: 8 cm)**

CHAPTER 11

11 Areas Related to Circles

Unit: Mensuration | 10 marks | Difficulty: Easy



★ SIMPLE IDEA

A sector is a **pizza slice**. θ tells you what fraction of the circle it is.

Σ FORMULAS

- Circumference = **$2\pi r$**, Area = **πr^2**
- **Arc length = $(\theta/360) \times 2\pi r$**
- **Sector area = $(\theta/360) \times \pi r^2$**
- Sector perimeter = arc + 2r
- Ring area = **$\pi(R^2 - r^2)$**
- Segment = sector − triangle

⚡ SHORTCUT TRICKS

- Use π = 22/7 when the radius is a **multiple of 7**.
- A clock's minute hand turns **6° every minute**.

✎ EXAMPLE

r = 7 cm, θ = 90°. Sector area?

90/360 = 1/4

1/4 × 22/7 × 49 = **38.5 cm^2**

⚠ DON'T DO THIS • MARKS BACHAO!

Use π = 22/7 **only if the question says so**. Write the units (cm^2).

✔ QUICK CHECK

Area of a circle with r = 7 cm? **(Answer: 154 cm^2)**

CHAPTER 12 • PART 1

12 Surface Areas & Volumes

Unit: Mensuration | 10 marks | Difficulty: Easy



★ SIMPLE IDEA

Surface area = **paint needed**. Volume = **water it can hold**.

◆ FORMULAS

Shape	Volume	Surface area
Cuboid	lbh	TSA = 2(lb + bh + hl), LSA = 2h(l + b)
Cube	a^3	TSA = $6a^2$, LSA = $4a^2$
Cylinder	$\pi r^2 h$	CSA = $2\pi rh$, TSA = $2\pi r(r + h)$
Cone	$\frac{1}{3}\pi r^2 h$	CSA = πrl, TSA = $\pi r(r + l)$

◆ EXTRAS

- **Cuboid diagonal** = $\sqrt{(l^2 + b^2 + h^2)}$
- **Cone slant height** $l = \sqrt{(r^2 + h^2)}$

⚡ TRICK

- A cone is exactly **one-third** of a cylinder with the same base and height.

⚠ DON'T DO THIS • MARKS BACHAO!

Convert all measurements to the **same unit** before you start.

✔ QUICK CHECK

Volume of a cube with side 3 cm? **(Answer: 27 cm^3)**

CHAPTER 12 • PART 2

12 Surface Areas & Volumes

Unit: Mensuration | 10 marks | Difficulty: Easy



Σ FORMULAS

- **Sphere:** Volume = 4/3 πr^3, Surface area = $4\pi r^2$
- **Hemisphere:** Volume = 2/3 πr^3, CSA = $2\pi r^2$, TSA = $3\pi r^2$
- **Frustum:** Volume = $\frac{1}{3}\pi h(r_1^2 + r_1 r_2 + r_2^2)$
- Frustum slant height: $l = \sqrt{(h^2 + (r_1 - r_2)^2)}$
- Frustum CSA = $\pi l(r_1 + r_2)$, TSA = CSA + $\pi r_1^2 + \pi r_2^2$

⚡ SHORTCUT TRICKS

- Melt and recast? The **volume stays the same**.
- Number of small objects = **big volume ÷ small volume**
- Hemisphere is half a sphere. TSA = curved part $2\pi r^2$ + flat top πr^2 = **$3\pi r^2$**
- **1 m^3 = 1000 litres**

✎ EXAMPLE

A sphere of radius 3 cm is melted into a cylinder of radius 1 cm. Height?

$4/3\ \pi(3)^3 = \pi(1)^2 h$

$36\pi = \pi h$

h = 36 cm

⚠ DON'T DO THIS • MARKS BACHAO!

In combined solids, **don't count the joined faces**. For hollow shapes, subtract the inner volume.

✔ QUICK CHECK

Volume of a hemisphere with r = 3 cm in terms of π? **(Answer: 18π cm^3)**

CHAPTER 13

13 Statistics

Unit: Statistics + Probability | 11 marks | Difficulty: Easy



★ SIMPLE IDEA

Mean is the average, **median** is the middle, **mode** is the most common.

Σ FORMULAS

- **Mean = Σfx / Σf**
- Assumed mean: **$\bar{x} = a + \Sigma fd / \Sigma f$**
- **Median = $l + [(n/2 - cf) / f] \times h$**
- **Mode = $l + [(f_1 - f_0) / (2f_1 - f_0 - f_2)] \times h$**
- **3 Median = Mode + 2 Mean**

◆ STEPS TO FIND THE MEDIAN

1. Make the **cumulative frequency (cf)** column
2. Find **n/2**
3. Median class = first class where cf **reaches or crosses** n/2
4. Put the values in the formula

Trick: Modal class = the class with the **highest frequency**.

✎ EXAMPLE

Mean = 30, Mode = 36. Median?

3 × Median = 36 + 2 × 30

3 × Median = 96

Median = 32

⚠ DON'T DO THIS • MARKS BACHAO!

“cf” is the cumulative frequency of the class **before** the median class.

✔ QUICK CHECK

Mean = 20, Median = 22. Mode? **(Answer: 26)**

CHAPTER 14

14 Probability

Unit: Statistics + Probability | 11 marks | Difficulty: Easy



★ SIMPLE IDEA

Chance = **what you want ÷ everything that can happen**.

Σ FORMULAS

- **P(E) = favourable outcomes / total outcomes**
- Always between **0 and 1**
- Sure event = **1**, impossible event = **0**
- **P(E) + P(not E) = 1**

◆ MUST-KNOW NUMBERS

- **Coin:** 2 outcomes. Two coins: 4. Three coins: 8
- **Die:** 6 outcomes. Two dice: 36
- **Cards:** 52 total, 26 red, 26 black, 12 face cards, 4 aces

✎ EXAMPLE

Probability of a prime number on a die

Primes: 2, 3, 5 → 3 favourable

Total = 6

P = 3/6 = **1/2**

⚠ DON'T DO THIS • MARKS BACHAO!

1 is NOT a prime number. Always simplify your fraction.

✔ QUICK CHECK

Probability of a king from a deck of 52? **(Answer: 1/13)**

BONUS • CHEAT SHEET

Formula Cheat Sheet (1/2)

Algebra | Geometry | Coordinate

Σ 1 REAL NUMBERS

- **HCF × LCM = Number 1 × Number 2** (two numbers only)
- HCF = common primes, **smallest** power
- LCM = all primes, **biggest** power
- Terminating decimal: denominator has only 2 and 5

Σ 2 POLYNOMIALS

- **Sum of zeros = −b/a**
- **Product of zeros = c/a**
- Polynomial: x^2 − (sum)x + (product)

Σ 3 PAIR OF LINEAR EQUATIONS

- $a_1/a_2 \neq b_1/b_2$ → **one** solution
- $a_1/a_2 = b_1/b_2 = c_1/c_2$ → **infinite**
- $a_1/a_2 = b_1/b_2 \neq c_1/c_2$ → **no** solution

Σ 7 COORDINATE GEOMETRY

- **Distance** = $\sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$
- **Midpoint** = $((x_1 + x_2)/2 , (y_1 + y_2)/2)$
- **Section** (m:n) = $((mx_2 + nx_1)/(m + n) , (my_2 + ny_1)/(m + n))$
- Area of triangle = ½ $|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$

Σ 4 QUADRATIC EQUATIONS

- **$D = b^2 - 4ac$**
- **$x = (-b \pm \sqrt{D}) / 2a$**
- D > 0: 2 different roots
- D = 0: 2 equal roots
- D < 0: no real roots

Σ 5 ARITHMETIC PROGRESSIONS

- **$d = a_2 - a_1$**
- **$a_n = a + (n - 1)d$**
- **$S_n = n/2 [2a + (n - 1)d]$**
- **$S_n = n/2 (a + l)$**
- 3 numbers: a − d, a, a + d

Σ 6 TRIANGLES

- Similarity: **AA, SSS, SAS**
- Area ratio = **(side ratio)^2**
- Pythagoras: $H^2 = B^2 + P^2$
- Triples: 3-4-5, 5-12-13, 8-15-17, 7-24-25

BONUS • CHEAT SHEET

Formula Cheat Sheet (2/2)

Trigonometry | Mensuration | Stats + Probability

Σ 8-9 TRIGONOMETRY

- **sin θ = P/H, cos θ = B/H, tan θ = P/B**
- cosec θ = H/P, sec θ = H/B, cot θ = B/P
- **$\sin^2\theta + \cos^2\theta = 1$**
- **$1 + \tan^2\theta = \sec^2\theta$**
- **$1 + \cot^2\theta = \text{cosec}^2\theta$**
- sin(90° − θ) = cos θ, tan(90° − θ) = cot θ
- sin (0° → 90°): 0, 1/2, 1/√2, √3/2, 1
- cos: same list in reverse
- tan: 0, 1/√3, 1, √3, not defined
- **tan θ = height / distance**

Σ 10 CIRCLES

- Tangent ⊥ radius at contact point
- Tangents from outside point are equal
- **Tangent length = $\sqrt{(d^2 - r^2)}$**

Σ 11 AREAS RELATED TO CIRCLES

- Circumference = **2πr**, Area = **πr^2**
- **Arc = (θ/360) × 2πr**
- **Sector = (θ/360) × πr^2**
- Ring = $\pi(R^2 - r^2)$
- Segment = sector − triangle

Σ 12 SURFACE AREAS & VOLUMES

- **Cuboid:** V = lbh, TSA = 2(lb + bh + hl)
- **Cube:** V = a^3, TSA = $6a^2$
- **Cylinder:** V = $\pi r^2 h$, CSA = 2πrh, TSA = 2πr(r + h)
- **Cone:** V = ⅓$\pi r^2 h$, CSA = πrl, TSA = πr(r + l), l = $\sqrt{(r^2 + h^2)}$
- **Sphere:** V = 4/3 πr^3, SA = $4\pi r^2$
- **Hemisphere:** V = 2/3 πr^3, CSA = $2\pi r^2$, TSA = $3\pi r^2$
- **Frustum:** V = ⅓$\pi h(r_1^2 + r_1 r_2 + r_2^2)$
- Frustum: CSA = $\pi l(r_1 + r_2)$, l = $\sqrt{(h^2 + (r_1 - r_2)^2)}$

Σ 13 STATISTICS

- **Mean = Σfx / Σf**
- **Median = $l + [(n/2 - cf)/f] \times h$**
- **Mode = $l + [(f_1 - f_0)/(2f_1 - f_0 - f_2)] \times h$**
- **3 Median = Mode + 2 Mean**

Σ 14 PROBABILITY

- **P(E) = favourable / total**
- 0 ≤ P(E) ≤ 1, P(E) + P(not E) = 1

BONUS • SPEED MATHS

Speed Table: Squares + Cubes

Calculation fast karo | Time bachao

Σ SQUARES 1 TO 25

$1^2 = \mathbf{1}$	$2^2 = \mathbf{4}$	$3^2 = \mathbf{9}$	$4^2 = \mathbf{16}$	$5^2 = \mathbf{25}$
$6^2 = \mathbf{36}$	$7^2 = \mathbf{49}$	$8^2 = \mathbf{64}$	$9^2 = \mathbf{81}$	$10^2 = \mathbf{100}$
$11^2 = \mathbf{121}$	$12^2 = \mathbf{144}$	$13^2 = \mathbf{169}$	$14^2 = \mathbf{196}$	$15^2 = \mathbf{225}$
$16^2 = \mathbf{256}$	$17^2 = \mathbf{289}$	$18^2 = \mathbf{324}$	$19^2 = \mathbf{361}$	$20^2 = \mathbf{400}$
$21^2 = \mathbf{441}$	$22^2 = \mathbf{484}$	$23^2 = \mathbf{529}$	$24^2 = \mathbf{576}$	$25^2 = \mathbf{625}$

Σ CUBES 1 TO 10

$1^3 = \mathbf{1}$	$2^3 = \mathbf{8}$	$3^3 = \mathbf{27}$	$4^3 = \mathbf{64}$	$5^3 = \mathbf{125}$
$6^3 = \mathbf{216}$	$7^3 = \mathbf{343}$	$8^3 = \mathbf{512}$	$9^3 = \mathbf{729}$	$10^3 = \mathbf{1000}$

★ YAAD RAKHO

- $\sqrt{2} \approx \mathbf{1.414}$, $\sqrt{3} \approx \mathbf{1.732}$, $\sqrt{5} \approx \mathbf{2.236}$
- $\pi \approx \mathbf{3.14}$ or $\mathbf{22/7}$ (only if the question says so)
- Primes up to 30: **2, 3, 5, 7, 11, 13, 17, 19, 23, 29**
- Right triangles: **3-4-5, 5-12-13, 8-15-17, 7-24-25**

EXAM-DAY PAGE

Exam-Day Page

80 marks | 3 hours | Last-minute revision

Σ PAPER PATTERN (80 MARKS, CHECK THE LATEST SAMPLE PAPER)

Section	Questions	Marks each
A	20 questions (MCQ + assertion-reason)	1
B	5 questions	2
C	6 questions	3
D	4 questions	5
E	3 case-based questions	4

⚠ TOP 10 MISTAKES THAT COST MARKS

1. Missing the minus sign in −b/a
2. No conclusion line in quadratics
3. Using HCF × LCM for three numbers
4. Swapping x and y
5. No reasons in geometry
6. Messy figure in trigonometry
7. Wrong value of π
8. No units
9. Calling 1 a prime number
10. Not checking the answer

◆ 3-HOUR PLAN

Time	What to do
15 min	Read the whole paper
90 min	Sections A, B, C
50 min	Sections D and E
25 min	Revision (check signs and units)

★ GOLDEN RULE

Always write the **formula first**, then substitute. You can earn **method marks** even if the final answer is wrong.

SELF-TEST

Self-Test + Answers

10 questions | Score yourself

✎ TRY THESE (NO PEEKING!)

1. HCF and LCM of 24 and 36

2. Zeros of $x^2 - 7x + 12$

3. Solve $x + y = 7$ and $x - y = 1$

4. Nature of roots of $x^2 - 4x + 4 = 0$

5. 10th term of AP 3, 7, 11...

6. Distance between (0, 0) and (6, 8)

7. If $\sin\theta = 5/13$, find $\cos\theta$

8. Tangent length from a point 10 cm from the centre, r = 6 cm

9. Volume of a sphere with r = 21 cm ($\pi = 22/7$)

10. Probability of a prime number on a die

✓ ANSWERS

1. 12 and 72

2. 3 and 4

3. x = 4, y = 3

4. Equal real roots (D = 0)

5. 39

6. 10

7. 12/13

8. 8 cm

9. 38,808 cm^3

10. 1/2

★ SCORE GUIDE

Score	Result
9 – 10	**Board-ready!**
6 – 8	Good. Revise your weak chapters.
Below 6	Read the chapters again and retry.

All the best for your boards! — Think Unlimited

BONUS • MOCK TEST

Mock Test (30 Minutes)

16 questions | 30 minutes | Answers next page

16 QUESTIONS, 16 MARKS

1. Find the HCF and LCM of 36 and 60. [Ch 1]

2. Find the zeros of $x^2 - 2x - 8$. Verify the sum and product. [Ch 2]

3. Solve: $2x + y = 11$ and $x - y = 1$. [Ch 3]

4. For what value of k does $2x^2 + kx + 8 = 0$ have equal roots? [Ch 4]

5. Find the 15th term of the AP 5, 8, 11, ... [Ch 5]

6. Find the sum of the first 20 terms of the AP 2, 4, 6, ... [Ch 5]

7. Two similar triangles have areas 16 cm^2 and 25 cm^2. Find the ratio of their corresponding sides. [Ch 6]

8. Find the distance of the point (3, 4) from the origin. [Ch 7]

9. If $\tan\theta = 3/4$ (θ is acute), find $\sin\theta$. [Ch 8]

10. Find the value of $\sin 60° \cos 30° + \sin 30° \cos 60°$. [Ch 8]

11. The angle of elevation of the top of a tower is 30° from a point $20\sqrt{3}$ m away. Find the height. [Ch 9]

12. A point is 17 cm from the centre of a circle of radius 8 cm. Find the tangent length. [Ch 10]

13. Find the area of a sector of radius 14 cm and angle 90° ($\pi = 22/7$). [Ch 11]

14. Find the volume of a cylinder with r = 7 cm and h = 10 cm ($\pi = 22/7$). [Ch 12]

15. Median = 45 and Mean = 42. Find the mode. [Ch 13]

16. One card is drawn from a deck of 52. Find P(a red face card). [Ch 14]

RULES

30 minutes. Book band. Formula pehle likho, phir substitute karo. Units likhna mat bhoolna!

BONUS • ANSWERS

Mock Test: Answers

Khud check karo | Mistakes note karo

✔ ANSWERS + WORKING

Q	Answer	Quick working
1	HCF = 12, LCM = 180	$36 = 2^2 \times 3^2$, $60 = 2^2 \times 3 \times 5$
2	4 and −2	$(x - 4)(x + 2) = 0$; sum = 2 = −b/a, product = −8 = c/a
3	x = 4, y = 3	Add the two equations: 3x = 12
4	k = ±8	$D = k^2 - 64 = 0$
5	47	5 + 14 × 3
6	420	$S_{20} = 20/2\ [4 + 19 \times 2] = 10 \times 42$
7	4 : 5	Side ratio = √(16/25)
8	5	$\sqrt{(3^2 + 4^2)}$
9	3/5	3-4-5 triangle: P = 3, H = 5
10	1	3/4 + 1/4
11	20 m	tan 30° = h / (20√3) → h = 20√3 / √3
12	15 cm	$\sqrt{(17^2 - 8^2)} = \sqrt{225}$
13	154 cm^2	1/4 × 22/7 × 14 × 14
14	1540 cm^3	22/7 × 7 × 7 × 10
15	51	Mode = 3 × 45 − 2 × 42
16	3/26	6 red face cards out of 52

★ SCORE GUIDE

Score	Result
14 – 16	**Board-ready!**
10 – 13	Good. Revise the chapters you missed.
Below 10	Read the chapters again, then retry.